

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. CHB-604: Physical Chemistry-IV

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 which is compulsory.

Question 1. Answer all parts of the following: [5 × 2 = 10]

- (a) Characterize the following molecules as linear, spherical tops, symmetrical tops, and asymmetrical tops: H₂O, CH₄, CH₃Cl, COS, C₆H₆.
- (b) Consider the three ensembles: microcanonical, canonical, and grand canonical. Which physical properties are constant in each of them?
- (c) What is zero-point energy? Explain.
- (d) State the physical significance of the partition function.
- (e) Differentiate between chemical shift and coupling constant in NMR.

Question 2.

- (a) Construct the Hamiltonian operator for a 1-D harmonic oscillator and set up the Schrödinger wave equation for this system. [6]
- (b) Discuss the Born-Oppenheimer approximation and its significance in molecular spectroscopy. [8]

Question 3.

- (a) Schematically represent the different vibrational modes of CO₂ and H₂O molecules. Identify which modes are IR-active. [6]
- (b) Show that the lines in the rotational spectrum of a diatomic molecule are equally spaced by $2B$. [8]

Question 4.

- (a) What is Larmor precession? Show that the Larmor precessional frequency is equal to the transition frequency between two nuclear spin states. [6]
- (b) State the Franck-Condon principle and explain its potential energy curve diagrams. [8]

Question 5. Answer any two of the following:

[2 × 7 = 14]

- (a) Postulates of Quantum Mechanics.
- (b) Derive an expression for the translational partition function of a molecule of mass m free to move in a 3-D box of volume V .
- (c) Ensembles and their thermodynamic relationships.